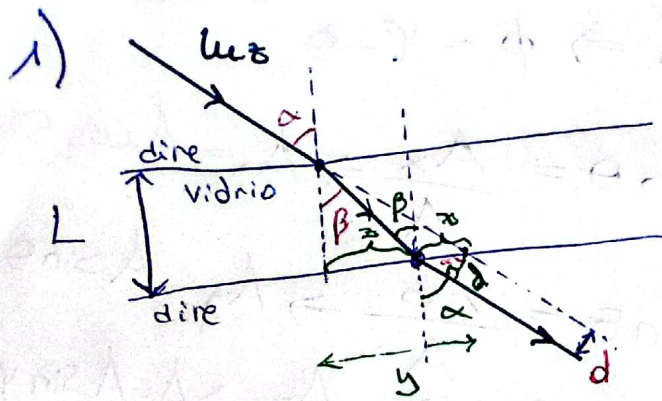




$$\alpha = 40^\circ$$

$$L = 2.0 \text{ cm}$$

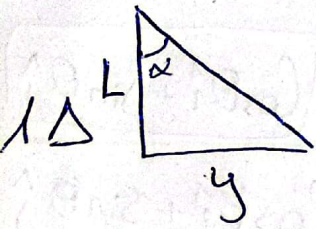
Ley de Snell

$$\frac{n_{\text{aire}}}{n_{\text{vidrio}}} \approx 1.5$$

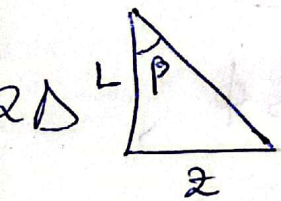
$$\frac{\sin \alpha}{\sin \beta} = \frac{n_{\text{aire}}}{n_{\text{vidrio}}} \approx 1.5$$

$$\beta = ? \Rightarrow \frac{\sin \alpha}{\sin \beta} \approx 1.5 \Rightarrow \sin \beta = \frac{\sin 40^\circ}{1.5}$$

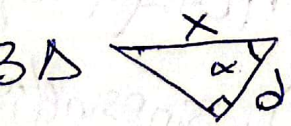
$$\Rightarrow \beta = \sin^{-1}\left(\frac{\sin 40^\circ}{1.5}\right) = 25.37^\circ$$



$$\tan 40^\circ = \frac{y}{2.0 \text{ cm}} \Rightarrow y = 2.0 \text{ cm} \tan 40^\circ = 1.68 \text{ cm}$$



$$\tan 25.37^\circ = \frac{z}{2.0 \text{ cm}} \Rightarrow z = 2.0 \text{ cm} \tan 25.37^\circ = 0.95 \text{ cm}$$



$$\text{Si } y = x + z \Rightarrow x = y - z = 1.68 \text{ cm} - 0.95 \text{ cm} = 0.73 \text{ cm}$$

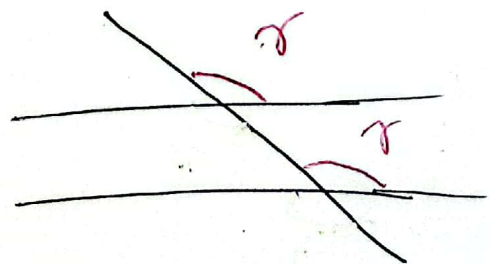
$$\cos 40^\circ = \frac{d}{x} \Rightarrow d = x \cos 40^\circ = (0.73 \text{ cm}) \cos 40^\circ$$

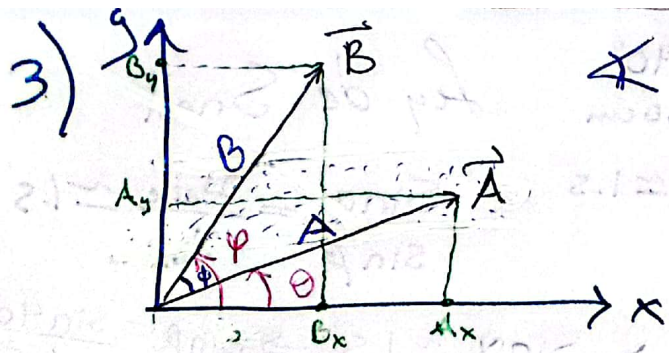
$$= \boxed{0.56 \text{ cm}}$$

$$2) \sin \alpha = \frac{\tan \alpha}{\sqrt{1 + \tan^2 \alpha}} = \frac{\sin \alpha}{\cos \alpha} \cdot \frac{1}{\sqrt{1 + \frac{\sin^2 \alpha}{\cos^2 \alpha}}}$$

$$= \frac{\sin \alpha}{\cos \alpha} \cdot \frac{1}{\sqrt{\frac{\cos^2 \alpha + \sin^2 \alpha}{\cos^2 \alpha}}} = \frac{\sin \alpha}{\cos \alpha} \cdot \frac{\cos \alpha}{\sqrt{\cos^2 \alpha + \sin^2 \alpha}}$$

$$= \frac{\sin \alpha}{\cancel{\cos \alpha}} \cdot \cancel{\cos \alpha} = \sin \alpha$$





$$\vec{A}, \vec{B} \Rightarrow \phi = \varphi - \theta$$

$$\cos \theta = \frac{A_x}{A} \Rightarrow A_x = A \cos \theta$$

$$\sin \theta = \frac{A_y}{A} \Rightarrow A_y = A \sin \theta$$

$$\cos \varphi = \frac{B_x}{B} \Rightarrow B_x = B \cos \varphi \quad \wedge \quad \sin \varphi = \frac{B_y}{B} \Rightarrow B_y = B \sin \varphi$$

$$\Rightarrow \text{Unitario } \vec{A}, \vec{B} \Rightarrow A=B=1$$

$$\Rightarrow \vec{A} = A_x \hat{i} + A_y \hat{j} = A \cos \theta \hat{i} + A \sin \theta \hat{j} = \boxed{\cos \theta \hat{i} + \sin \theta \hat{j}}$$

$$\Rightarrow \vec{B} = B_x \hat{i} + B_y \hat{j} = B \cos \varphi \hat{i} + B \sin \varphi \hat{j} = \boxed{\cos \varphi \hat{i} + \sin \varphi \hat{j}}$$

Producto Punto $\vec{A} \cdot \vec{B} = |A||B| \cos \phi$

$$\Rightarrow (\cos \theta \hat{i} + \sin \theta \hat{j}) \cdot (\cos \varphi \hat{i} + \sin \varphi \hat{j}) = |1||1| \cos(\varphi - \theta)$$

$$\Rightarrow \cos \theta \cos \varphi (\hat{i} \cdot \hat{i}) + \cos \theta \sin \varphi (\hat{i} \cdot \hat{j}) + \sin \theta \cos \varphi (\hat{j} \cdot \hat{i}) + \sin \theta \sin \varphi (\hat{j} \cdot \hat{j}) = \cos(\varphi - \theta)$$

$$\Rightarrow \boxed{\cos \theta \cos \varphi + \sin \theta \sin \varphi = \cos(\varphi - \theta)}$$